DC Circuits — Reference

Textbook: Ulaby, Maharbiz, Furse, *Circuit Analysis and Design*, 3rd ed., chapters 1–2. Gray tags like [book 2-1] point to the book section. The book writes voltage as υ (upsilon). These notes write v.

prefixes

Symbol	Prefix	Power	Multiply by	In words
E	exa	10^{18}		quintillion
P	peta	10^{15}		quadrillion
T	tera	10^{12}	1,000,000,000,000	trillion
G	giga	10^{9}	1,000,000,000	billion
M	mega	10^{6}	1,000,000	million
k	kilo	10^{3}	1,000	thousand
—	(none)	10^{0}	1	one
m	milli	10^{-3}	0.001	thousandth
μ	micro	10^{-6}	0.000 001	millionth
n	nano	10^{-9}	0.000 000 001	billionth
p	pico	10^{-12}	0.000 000 000 001	trillionth
f	femto	10^{-15}		quadrillionth
a	atto	10^{-18}		quintillionth

Capital M (mega) and small m (milli) are 10^{9} apart. [book Table 1-2]

quantities and units

Quantity	Symbol	Unit	In base units
charge	q, Q	coulomb (C)	A·s
current	i, I	ampere (A)	C/s
voltage	v, V (book: υ)	volt (V)	J/C
power	p, P	watt (W)	J/s = V·A
energy	w, W	joule (J)	W·s
resistance	R	ohm (Ω)	V/A
conductance	G	siemens (S)	A/V = $1/\Omega$
resistivity	ρ (rho)	ohm-meter	Ω·m
conductivity	σ (sigma)	siemens/meter	S/m
capacitance	C	farad (F)	C/V
inductance	L	henry (H)	V·s/A
frequency	f	hertz (Hz)	1/s
length, area	ℓ, A	m, m^2	—
time	t	second (s)	—
temperature	T	kelvin (K), °C	K = °C+273

Lowercase (q, i, v, p, w) can change with time. Uppercase (Q, I, V, P, W) is a constant (dc). [book Table 1-1]

constants and conversions

Item	Value
charge of one electron	$-e$, $e = 1.6 \times 10^{-19}$ C (more exact: 1.602)
charge of one proton	$+e$
electrons in 1 C	6.25×10^{18}
minute, hour	1 min = 60 s, 1 h = 3600 s
ampere-hour	1 Ah = 3600 C, 1 mAh = 3.6 C
watt-hour, kilowatt-hour	1 Wh = 3600 J, 1 kWh = 3.6×10^{6} J
length	1 mm = 10^{-3} m, 1 cm = 10^{-2} m
area	1 mm^2 = 10^{-6} m^2, 1 cm^2 = 10^{-4} m^2
specific heat of water	$c = 4186$ J/(kg·°C)
mass of water	1 L ≈ 1 kg, 1 fl oz ≈ 0.02957 kg

The book uses $e = 1.6 \times 10^{-19}$ C. Use it to match the book's answers.

circuit terminology

Term	Meaning [book Table 1-4]
node	all the wire between elements: one voltage. Points joined by a wire or a short are the same node
ordinary node	connects to 2 elements
extraordinary node	connects to 3 or more elements
branch	one element between two nodes
path	a chain of branches that repeats no node
extraordinary path	a path between two adjacent extraordinary nodes
loop	a closed path
independent loop	a loop with at least one branch that no other chosen loop has
mesh	a loop that encloses no other loop
in series	same current: only ordinary nodes between them
in parallel	same voltage: they share both end nodes
planar circuit	can be drawn with no crossing wires. A crossing without a dot is not a connection
open circuit	$i = 0$, $R = \infty$, any voltage
short circuit	$v = 0$, $R = 0$, any current
ground	the 0 V reference node. Chassis ground: the metal frame. Earth ground: a rod in the soil

conductivity and resistivity, 20°C

Material	σ (S/m)	$\rho = 1/\sigma$ (Ω·m)
Conductors		
silver	6.17×10^{7}	1.62×10^{-8}
copper	5.81×10^{7}	1.72×10^{-8}
gold	4.10×10^{7}	2.44×10^{-8}
aluminum	3.82×10^{7}	2.62×10^{-8}
iron	1.03×10^{7}	9.71×10^{-8}
mercury (liquid)	1.04×10^{6}	9.58×10^{-7}
Semiconductors		
carbon (graphite)	7.14×10^{4}	1.40×10^{-5}
pure germanium	2.13	0.47
pure silicon	4.35×10^{-4}	2.30×10^{3}
Insulators		
paper	$\sim 10^{-10}$	$\sim 10^{10}$
glass	$\sim 10^{-12}$	$\sim 10^{12}$
teflon	$\sim 3.3 \times 10^{-13}$	$\sim 3 \times 10^{12}$
porcelain	$\sim 10^{-14}$	$\sim 10^{14}$
mica	$\sim 10^{-15}$	$\sim 10^{15}$
polystyrene	$\sim 10^{-16}$	$\sim 10^{16}$
fused quartz	$\sim 10^{-17}$	$\sim 10^{17}$
Common materials		
distilled water	5.5×10^{-6}	1.8×10^{5}
drinking water	$\sim 5 \times 10^{-3}$	~ 200
sea water	4.8	0.2
rubber	10^{-13}	10^{13}
Biological tissues		
blood, muscle	~ 1.5	~ 0.67
fat	~ 0.1	10

Small ρ (big σ) = good conductor. [book Table 2-1] If a problem gives ρ or σ, use the problem's value.

wire size (AWG) and cross-section area

AWG	Diameter d (mm)	Area $A = \pi d^2/4$ (m^2)
0	8.3	5.41×10^{-5}
2	6.5	3.32×10^{-5}
4	5.2	2.12×10^{-5}
6	4.1	1.32×10^{-5}
10	2.6	5.31×10^{-6}
14	1.6	2.01×10^{-6}
18	1.0	7.85×10^{-7}
20	0.8	5.03×10^{-7}

Bigger AWG number = thinner wire. [book Table 2-2]

Formula	In words
$A = \pi r^2 = \dfrac{\pi d^2}{4}$	round wire: (area) $= \pi$(radius)^2
$d = \sqrt{\dfrac{4A}{\pi}}$	(diameter) from the area
$A = w \cdot h$	rectangle: (width)(thickness), the face the current passes *through*
$A = \pi(b^2 - a^2)$	hollow tube: outer radius b, inner radius a

Convert mm to m before you square: $(1\text{ mm})^2 = 10^{-6}\text{ m}^2$. ℓ is along the current, A is across it. For a block, the current direction decides which side is ℓ.

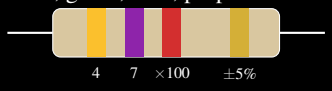
resistor color code

	Color	Digit	Multiplier	Tolerance	Temp. coef.
	black	0	$\times 1$	—	—
	brown	1	$\times 10$	$\pm 1\%$	100 ppm
	red	2	$\times 100$	$\pm 2\%$	50 ppm
	orange	3	$\times 1$k	—	15 ppm
	yellow	4	$\times 10$k	—	25 ppm
	green	5	$\times 100$k	$\pm 0.5\%$	—
	blue	6	$\times 1$M	$\pm 0.25\%$	10 ppm
	purple (violet)	7	$\times 10$M	$\pm 0.1\%$	5 ppm
	gray	8	—	—	—
	white	9	—	—	—
	gold	—	$\times 0.1$	$\pm 5\%$	—
	silver	—	$\times 0.01$	$\pm 10\%$	—

[book Fig. 2-2] Hold it so the wider gap is before the last band(s), on the right. Read left to right.

Formula	In words
4 bands: $R = (b_1 b_2) \times 10^{b_4} \pm b_5$	(two digits) $\times$ (multiplier) $\pm$ (tolerance)
5 bands: $R = (b_1 b_2 b_3) \times 10^{b_4} \pm b_5$	(three digits) $\times$ (multiplier) $\pm$ (tolerance)
6 bands: add b_6	5-band reading plus the temperature coefficient, ppm/°C

The multiplier band is the number of zeros. Digits 2–7 follow the rainbow: red, orange, yellow, green, blue, purple.



Quick example	Work → answer
yellow, purple, red, gold (above)	$47 \times 100 = 4.7$ kΩ, ±5%: 4465 to 4935 Ω
brown, black, orange, gold	$10 \times 10^3 = 10$ kΩ, ±5%
5 bands: brown, black, black, red, brown	$100 \times 10^2 = 10$ kΩ, ±1%
red, red, gold, gold	$22 \times 0.1 = 2.2\,\Omega$, ±5%
(reverse) 330 Ω, ±5%	33×10^1: orange, orange, brown, gold

formula sheet (symbols only)

2. charge, current	$Q = It$, $i = \dfrac{dq}{dt}$, $\Delta Q = \int_{t_1}^{t_2} i\,dt$, $n = \dfrac{\lvert Q\rvert}{e}$
3. voltage	$v_{ab} = \dfrac{dw}{dq}$, $V_{ab} = V_a - V_b$, $W = QV$
4. power, energy	$p = vi$, $\sum p = 0$, $w = \int p\,dt$, $W = Pt$, $P_{av} = \dfrac{1}{T}\int_0^T p\,dt$
5. extras	$\text{Wh} = V \cdot \text{Ah}$, $t = \dfrac{\text{Ah}}{I}$, $E = cm\Delta T$, $\text{kWh} = \dfrac{\text{J}}{3.6 \times 10^6}$
6. sources	$V_L = \dfrac{R_L}{R_s + R_L} v_s$; dependent: αv_x, $r i_x$, $g v_x$, βi_x
7. resistors	$V = IR$, $R = \dfrac{\rho \ell}{A} = \dfrac{\ell}{\sigma A}$, $P = I^2R = \dfrac{V^2}{R}$, $R = R_0(1 + \alpha T)$
8. KCL, KVL	$\sum i_{\text{in}} = \sum i_{\text{out}}$, $\sum v = 0$; N unknowns: M KVL + $(N - M)$ KCL
9. combine	$R_1 + R_2$, $\dfrac{R_1 R_2}{R_1 + R_2}$, $G_{eq} = \sum G$
10. dividers	$v_1 = \dfrac{R_1}{R_1 + R_2} v_s$, $i_1 = \dfrac{R_2}{R_1 + R_2} i_s = \dfrac{G_1}{G_{eq}} i_s$
11. source transf.	$i_s = \dfrac{v_s}{R_s}$, $v_s = i_s R_s$ (same R_s)
12. $\Delta \to Y$	$R_1 = \dfrac{R_b R_c}{R_a + R_b + R_c}$, $R_a = \dfrac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$
13. bridge	$R_x = \dfrac{R_2}{R_1} R_3$, $V_{out} \approx \dfrac{V_0}{4} \dfrac{\Delta R}{R}$, $R = R_0(1 + \alpha P)$
14. diode, LED	forward: $I = \dfrac{V_s - V_F}{R}$, $V_F \approx 0.7$ V (Si)

calculus and algebra tools

Derivatives (a, ω, n, c constants):

Function	Derivative
c, t^n	0, nt^{n-1}
e^{at}	$a e^{at}$
$\sin(\omega t)$, $\cos(\omega t)$	$\omega \cos(\omega t)$, $-\omega \sin(\omega t)$
$f \cdot g$ (product)	$f'g + fg'$
$f(g(t))$ (chain)	$f'(g) \cdot g'$; e.g. $\sin^2(\omega t) \to 2\omega \sin(\omega t)\cos(\omega t)$

Integrals:

Function	Antiderivative
t^n	$\dfrac{t^{n+1}}{n+1}$
e^{at}	$\dfrac{e^{at}}{a}$
$\sin(\omega t)$, $\cos(\omega t)$	$-\dfrac{\cos(\omega t)}{\omega}$, $\dfrac{\sin(\omega t)}{\omega}$
$e^{at}\cos(bt)$	$\dfrac{e^{at}(a\cos bt + b\sin bt)}{a^2 + b^2}$
$e^{at}\sin(bt)$	$\dfrac{e^{at}(a\sin bt - b\cos bt)}{a^2 + b^2}$
$\int_0^\infty Ae^{-at}\,dt$	$\dfrac{A}{a}$ $(a > 0)$

Definite integral: $F(\text{top}) - F(\text{bottom})$. $e^0 = 1$. $e^{-at} \to 0$ as $t \to \infty$.
Trig identities: $\sin x \cos x = \frac{1}{2}\sin 2x$, $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$, $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$.
Maximum or minimum: set the derivative to 0 and solve for t. Example: largest q where $\frac{dq}{dt} = 0$; largest p where $\frac{dp}{dt} = 0$.

Solving simultaneous equations

2 unknowns ($a_1 x + b_1 y = c_1$, $a_2 x + b_2 y = c_2$), Cramer's rule:

$$D = a_1 b_2 - a_2 b_1 \qquad x = \frac{c_1 b_2 - c_2 b_1}{D} \qquad y = \frac{a_1 c_2 - a_2 c_1}{D}$$

3 unknowns:

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1(b_2 c_3 - b_3 c_2) - b_1(a_2 c_3 - a_3 c_2) + c_1(a_2 b_3 - a_3 b_2).$$

$x_k = \dfrac{D_k}{D}$, where D_k is D with column k replaced by the right-hand sides. Or: eliminate one unknown at a time, or use a calculator's matrix solver (rref).

[book App. B]

Quick example	Work → answer
$3x+2y=12,\quad x-y=-1$	$D=3(-1)-1(2)=-5;\quad x=\frac{12(-1)-(-1)(2)}{-5}=2,$ $y=\frac{3(-1)-1(12)}{-5}=3$

which tool? — given → find

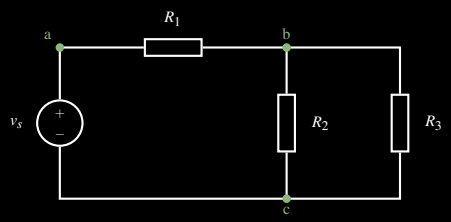
The number in parentheses is the concept section.

Given → find	Do this	
a circuit → nodes, series, parallel	terminology table	(1)
charge Q → electrons	$n=\lvert Q\rvert/e$	(2)
$q(t)\to i$; $\quad i(t)\to\Delta Q$	differentiate; integrate	(2)
a $q(t)$ or $i(t)$ graph	slope; area	(2)
node voltages, new ground	subtract the new ground's old value	(3)
V, I → power, sign	$P=VI$, passive sign convention	(4)
$v(t)$, $i(t)$ → energy, average power	$\int vi\,dt$; $\quad\frac{1}{T}\int p\,dt$	(4)
battery, kWh, heat, cost	extras	(5)
switch at t_0	redraw: closed = short, open = open	(6)
two of V, I, R → the third	Ohm's law	(7)
wire: $R\leftrightarrow\ell$, A, material	$R=\rho\ell/A$ and the tables	(7)
resistance at a new temperature	$R=R_0(1+\alpha T)$	(7)
color bands → R	color chart (ref)	
currents at a node; voltages in a loop	KCL; KVL	(8)
N unknown currents	M KVL + $(N-M)$ KCL, solve	(8)
network → R_{eq}	combination chart	(9)
series → v share; parallel → i share	dividers	(10)
mixed sources and resistors	source transformation	(11)
not series, not parallel (a bridge)	$\Delta\leftrightarrow Y$	(12)
bridge balance; sensor output	Wheatstone	(13)
force on a sensor → V_{out}	piezoresistor, cantilever	(13)
diode or LED in a circuit	assume on, check $I>0$	(14)

DC Circuits — Concepts

1. reading a circuit [book 1-3]

Label the nodes first. Every wire segment between elements is one node, even if it is drawn long. Merge points joined by a wire or a short.



Quick example	Work → answer
nodes above	a (ordinary: v_s, R_1), b and c (extraordinary: 3 elements each)
series pairs above	v_s and R_1: only ordinary node a between them
parallel pairs above	R_2 and R_3: both connect to b and c
meshes above	2: (v_s, R_1, R_2) and (R_2, R_3)

Meters [book Fig. 1-18]: a voltmeter goes in parallel (across the element), red lead on $+$. Ideal: draws no current. An ammeter goes in series (in the path). Ideal: no voltage drop.
Same branch, same direction = same current. Two currents in one branch with opposite arrows are negatives.

2. charge and current [book 1-4]

Current is the flow of positive charge. Electrons ($-e$) moving one way count as current the other way. Protons ($+e$) count in their own direction.

Formula	In words
$n = \dfrac{\lvert Q\rvert}{e}$, $Q = ne$	(electrons) = (charge) ÷ (charge of one)
$Q = It$, $I = \dfrac{Q}{t}$, $t = \dfrac{Q}{I}$	(charge) = (current)(time), constant current
$i(t) = \dfrac{dq}{dt}$	(current) = rate of change of (charge): differentiate
$q(t) = \int_{-\infty}^{t} i\,dt$	(charge so far) = total of (current): integrate
$\Delta Q = q(t_2) - q(t_1) = \int_{t_1}^{t_2} i\,dt$	(charge moved from t_1 to t_2)
graph of $q(t)$	current = slope of each straight piece
graph of $i(t)$	charge = area under it (piece by piece)

Quick example	Work → answer
2 mC → electrons	$\dfrac{2\times10^{-3}}{1.6\times10^{-19}} = 1.25\times10^{16}$
3×10^{18} electrons/s go $+x$; 10^{18} protons/s go $-x$	both are current toward $-x$: $4\times10^{18}\cdot1.6\times10^{-19} = 0.64$ A in $-x$
90 C in 30 s; 0.5 A for 2 min	3 A; $0.5\cdot120 = 60$ C
$q = 3t^2$ mC → i	$6t$ mA
$q = 0.8(1-e^{-0.5t})$ nC → i	$0.8\cdot0.5e^{-0.5t} = 0.4e^{-0.5t}$ nA
q rises 0 → 6 C in a line over 0–3 s	slope: $6/3 = 2$ A
$i = 4$ A for 2 s, then 0	area: $4\cdot2 = 8$ C
$q = 10te^{-t}$ C, when is q largest?	$\frac{dq}{dt} = 10e^{-t}(1-t) = 0 \Rightarrow t = 1$ s, $q = 10/e = 3.68$ C
$i = 0.4e^{-0.5t}$ nA, total to ∞	$\frac{A}{a} = \frac{0.4}{0.5} = 0.8$ nC

Time in seconds. Given charge, find current: differentiate. If your answer has a higher power of t, you integrated.

Worked: (integrate) $i(t) = 20e^{-0.5t}$ mA from $t = 0$ to 4 s

1. ANTIDERIVATIVE
$F(t) = \dfrac{20}{-0.5}e^{-0.5t} = -40e^{-0.5t}$

↓

2. TOP ($t = 4$)
$F(4) = -40e^{-2} = -40(0.1353) = -5.413$

↓

3. BOTTOM ($t = 0$)
$F(0) = -40e^{0} = -40$ $(e^0 = 1)$

↓

4. TOP MINUS BOTTOM
$-5.413 - (-40) = 34.59$ mC $(mA\cdot s = mC)$

Worked: (product rule) $q(t) = 0.5t\cos(100\pi t)$ μC

1. PARTS
$f = 0.5t$, $f' = 0.5$. $g = \cos(100\pi t)$, $g' = -100\pi\sin(100\pi t)$

↓

2. $f'g + fg'$
$i = 0.5\cos(100\pi t) - 50\pi t\sin(100\pi t)$ μA

3. voltage, node voltage, and ground [book 1-5.1]

Voltage is energy per charge between two points. It is *across* an element. A **node voltage** is measured from ground (0 V).

Formula	In words
$v_{ab} = \dfrac{dw}{dq}$	(voltage of a over b) = (energy) per (charge)
$V = \dfrac{W}{Q}$, $W = QV$	constant case
$V_{ab} = V_a - V_b$	(voltage between two nodes) = (node voltage a) − (node voltage b)
$V_{ab} = -V_{ba}$	reversing the order flips the sign

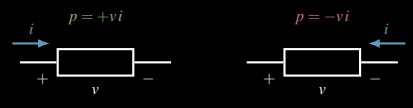
Moving ground shifts every node voltage by the same amount. Voltage *differences* do not change.

Quick example	Work → answer
30 J moves 5 C	$V = 30/5 = 6$ V
$V_a = 9$, $V_b = 5$, $V_c = 0$ (ground c) → ground at b	subtract 5: $V_a = 4$, $V_b = 0$, $V_c = -5$ V
V_{ab} before and after	$9-5 = 4$ and $4-0 = 4$: same

4. power and energy [book 1-5.2]

Formula	In words
$p = v\,i$	(power) = (voltage)(current)
$I = \dfrac{P}{V}$, $V = \dfrac{P}{I}$	rearranged
$\sum_k p_k = 0$	conservation: (power supplied) = (power received)
$w = \int_{t_1}^{t_2} p\,dt$	(energy) = total of (power) over time
$W = Pt$, $P = \dfrac{W}{t}$, $t = \dfrac{W}{P}$	constant power
$W = V\,I\,t$	(energy) = (voltage)(current)(time)
$P_{av} = \dfrac{1}{T}\int_0^T p\,dt$	(average power) = (energy over T) ÷ T

Passive sign convention: i enters the $+$ side of $v \Rightarrow p = +vi$. Enters the $-$ side $\Rightarrow p = -vi$. Then $p > 0$: the device receives (absorbs) power. $p < 0$: it supplies power.



Quick example	Work → answer
5 V, 2 A into +; into −	+10 W receives; −10 W supplies
−3 V, 4 A into +	$+(-3)(4) = -12$ W: supplies
three devices: +20, −50, p_3	$20 - 50 + p_3 = 0 \Rightarrow p_3 = 30$ W
24 W for 1 h	$24 \cdot 3600 = 86{,}400$ J
$p = 10t$ W over 0–2 s → P_{av}	$\frac{1}{2}\int_0^2 10t\,dt = \frac{1}{2}(20) = 10$ W
$p(t)$ graph	energy = area under p; build $p = v \cdot i$ piece by piece
$v = 50(1 - e^{-0.5t})$ V, $i = 4e^{-0.5t}$ mA at $t = 1$	$19.67 \cdot 2.426 = 47.7$ mW

$\int v\,i\,dt \neq \int v\,dt \cdot \int i\,dt$. Multiply first. Maximum power: set $\frac{dp}{dt} = 0$.

***Worked: total energy,* $v = 50(1 - e^{-0.5t})$ *V,* $i = 4e^{-0.5t}$ *mA,* $0 \to \infty$**

1. MULTIPLY
$p = 200\left(1 - e^{-0.5t}\right)e^{-0.5t}$ mW

↓

2. DISTRIBUTE, COMBINE EXPONENTS
$p = 200e^{-0.5t} - 200e^{-t}$ $\quad (e^{-0.5t}e^{-0.5t} = e^{-t})$

↓

3. ANTIDERIVATIVE OF EACH TERM
$F(t) = -400e^{-0.5t} + 200e^{-t}$

↓

4. TOP ($t \to \infty$) AND BOTTOM ($t = 0$)
Top: 0. Bottom: $-400 + 200 = -200$

↓

5. TOP MINUS BOTTOM
$0 - (-200) = 200$ mJ

5. batteries, utility energy, heat, cost

Course-assignment topics. Ah is charge; Wh and kWh are energy.

Formula	In words
$\text{Wh} = V \cdot \text{Ah}$	(energy rating) = (voltage)(charge rating)
$t = \frac{\text{Ah}}{I} = \frac{\text{Wh}}{P}$	(run time, hours)
$\text{J} = \text{Wh} \cdot 3600$, $\text{C} = \text{Ah} \cdot 3600$	hours to seconds
$n = \frac{W}{W_{\text{one}}}$	(number of batteries filled), round down
$E = cm\Delta T$	(heat) = (specific heat)(mass)(temperature change)
$\Delta T = \frac{E}{cm}$, $m = \frac{E}{c\Delta T}$	rearranged
$\text{cost} = \frac{\text{J}}{3.6 \times 10^6} \cdot \text{rate}$	joules → kWh → cost

Quick example	Work → answer
2.5 Ah, 500 mA load	$2.5/0.5 = 5$ h
12 V, 60 Wh, 250 mA load	$P = 3$ W; $60/3 = 20$ h
1.08 MJ fills 3000 mAh, 3.7 V batteries	one: 11.1 Wh = 39,960 J; $27.03 \to 27$
0.5 kg water, 20 → 80°C	$4186 \cdot 0.5 \cdot 60 = 125{,}580$ J
125,580 J at 15 cents/kWh	$0.03488 \cdot 15 = 0.523$ cents

6. circuit elements, sources, switches [book 1-6]

Active devices need a power source (transistors, ICs). **Passive** devices do not (resistors, capacitors, inductors).

Element	Rule
ideal voltage source	$v = V_s$ for any current. i–v graph: vertical line
ideal current source	$i = I_s$ for any voltage. i–v graph: horizontal line
real voltage source	v_s in series with R_s. Ignore R_s if $R_s \ll R_L$
real current source	i_s in parallel with R_s. Ignore R_s if $R_s \gg R_L$
SPST switch	closed = short, open = open. The label $t = t_0$ is when it changes
SPDT switch	the common terminal moves from contact 1 to contact 2 at t_0

Formula	In words
$V_L = \frac{R_L}{R_s + R_L} v_s$	(load voltage) of a real source
$v = \alpha v_x$	VCVS: α has no units
$v = r i_x$	CCVS: r in V/A = Ω
$i = g v_x$	VCCS: g in A/V = S
$i = \beta i_x$	CCCS: β has no units

Quick example	Work → answer
9 V, $R_s = 1\,\Omega$, $R_L = 8\,\Omega$	$V_L = \frac{8}{9} \cdot 9 = 8$ V
CCVS $10 i_x$, $i_x = 0.5$ A	5 V
VCCS $0.2 v_x$, $v_x = 10$ V	2 A
switch closes at $t = 2$ s	two circuits: $t < 2$ (open branch), $t > 2$ (short)

Dependent sources are models, not energy sources. Solve them like any source: their value comes from v_x or i_x.

7. Ohm's law and resistance [book 2-1]

Formula	In words
$V = IR$	(voltage) = (current)(resistance)
$I = \frac{V}{R}$, $R = \frac{V}{I}$	rearranged
$I = \frac{V_a - V_b}{R}$	current from a to b through R
$G = \frac{1}{R}$, $I = GV$	(conductance) = 1÷(resistance)

Ohm's law follows the passive sign convention. If I enters the − side, $I = -V/R$.

Resistance of a shape:

Formula	In words
$R = \frac{\rho\ell}{A} = \frac{\ell}{\sigma A}$	(resistance) = (resistivity)(length) ÷ (area)
$\ell = \frac{RA}{\rho} = R\sigma A$	(length) = (resistance)(area) ÷ (resistivity)
$A = \frac{\rho\ell}{R}$	(area) = (resistivity)(length) ÷ (resistance)
$\rho = \frac{RA}{\ell}$, $\sigma = \frac{\ell}{RA}$	identify the material from the table

Temperature [book P2.5]:

Formula	In words
$R = R_0(1 + \alpha T)$	R_0 at 0°C, T in °C, α in $^\circ\text{C}^{-1}$
$T = \frac{R/R_0 - 1}{\alpha}$	(temperature) for a given R
$\frac{R_2}{R_1} = \frac{1 + \alpha T_2}{1 + \alpha T_1}$	compare two temperatures

Copper: $\alpha = 4 \times 10^{-3}$ $^\circ\text{C}^{-1}$. A **thermistor** is built to change a lot with T: $\Delta R = k\Delta T$.

Power and power rating:

Formula	In words
$P = VI = I^2R = \frac{V^2}{R} = GV^2$	power in a resistor (always received, becomes heat)
$R = \frac{V^2}{P}$, $R = \frac{P}{I^2}$	(resistance) from power
$I_{max} = \sqrt{\frac{P_{rated}}{R}}$, $V_{max} = \sqrt{P_{rated}R}$	largest safe current, voltage
$\text{fraction} = \frac{P_{load}}{P_{source}} = \frac{I^2R_L}{VI}$	share of the power that reaches the load

Variable resistors [book Table 2-3]: a **rheostat** (2 terminals) sets 0 to R_{max}. A **potentiometer** (3 terminals) splits R_{max}: $R_{13} + R_{23} = R_{max}$.

Quick example	Work → answer
10 V across 2 kΩ	$I = 5$ mA
100 W lamp at 120 V	$R = 120^2/100 = 144\,\Omega$
2 W in 50 Ω	$I = \sqrt{2/50} = 0.2$ A
$\frac{1}{4}$ W rating, 1 kΩ	$I_{max} = \sqrt{0.25/1000} = 15.8$ mA
carbon film: $\ell = 2$ cm, 1 mm wide, 0.1 mm thick	$A = 10^{-3}\cdot 10^{-4} = 10^{-7}$ m^2; $R = \frac{1.40\times10^{-5}\cdot 0.02}{10^{-7}} = 2.8\,\Omega$
twin cable 10 m long, AWG-14 copper	both wires: $\ell = 20$ m; $R = \frac{1.72\times10^{-8}\cdot 20}{2.01\times10^{-6}} = 0.171\,\Omega$
12 V, cable 0.1 Ω, load 2.9 Ω	$I = 4$ A; $\frac{16\cdot 2.9}{12\cdot 4} = \frac{46.4}{48} = 0.967$
copper $R_0 = 100\,\Omega$ at 50°C	$100(1 + 0.004\cdot 50) = 120\,\Omega$
when is copper R 10% above R_0?	$T = \frac{1.1-1}{0.004} = 25$°C
10 kΩ pot, wiper 30% from terminal 1	$R_{13} = 3$ kΩ, $R_{23} = 7$ kΩ

Wire design chain	Copper, AWG-18, target $R = 0.5\,\Omega$
1. R (from V, I if given)	$R = \frac{V}{I}$; here 0.5 Ω is given
2. d from the AWG table	1.0 mm $= 1.0\times10^{-3}$ m
3. area	$A = \frac{\pi(1.0\times10^{-3})^2}{4} = 7.854\times10^{-7}$ m^2
4. length	$\ell = \frac{RA}{\rho} = \frac{0.5\cdot 7.854\times10^{-7}}{1.72\times10^{-8}} = 22.8$ m

A two-wire cable is twice as long as the run. Current goes out on one wire and back on the other.

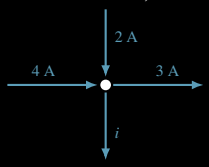
8. Kirchhoff's laws [book 2-2]

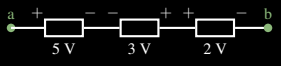
Formula	In words
$\sum i_{in} = \sum i_{out}$	KCL: (current into a node) = (current out)
$\sum_n i_n = 0$	KCL, entering = +, leaving = − (or the reverse)
$\sum_n v_n = 0$	KVL: voltages around a closed loop add to 0
$\sum v_{rises} = \sum v_{drops}$	KVL, other form
V_{ab} = sum along any path a→b	path rule: $+v$ if you meet + first

KVL sign rule: go clockwise. Enter an element at +: write $+v$. Enter at −: write $-v$. KCL also holds for a closed boundary around several nodes.

How many equations? [book 2-2.3]

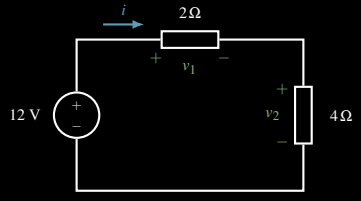
1. Count the unknowns: N.
2. Write M KVL equations. Skip loops through a current source. Each new loop must use an element not used before.
3. Write $N - M$ KCL equations at extraordinary nodes.
4. Put Ohm's law in, then solve the system (Cramer's rule, reference page).





Quick example	Work → answer
node (left figure)	$4 + 2 = 3 + i \Rightarrow i = 3$ A
1 A in; 3 A and i out	$i = -2$: 2 A in, against the arrow
10 A in; i_x and $1.5\,i_x$ out	$10 = 2.5\,i_x \Rightarrow i_x = 4$ A
V_{ab} (right figure)	$+5 - 3 + 2 = 4$ V
12 V source, 5 V on R_1, find v_2	$-12 + 5 + v_2 = 0 \Rightarrow 7$ V
36 V, 2 Ω, 4 Ω, CCVS $3i$ in one loop	$36 = 2i + 4i + 3i \Rightarrow i = 4$ A
3 unknown currents, 1 current source	M = loops that avoid the source; rest KCL

A negative answer is fine. It means the current flows opposite to your arrow.



Worked: KVL around the loop above

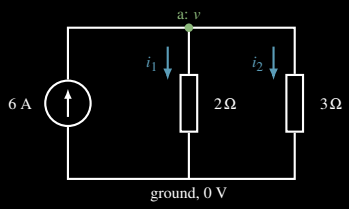
1. KVL CLOCKWISE
$-12 + v_1 + v_2 = 0$

↓

2. OHM'S LAW IN
$-12 + 2i + 4i = 0 \Rightarrow 6i = 12 \Rightarrow i = 2$ A

↓

3. VOLTAGES AND POWER
$v_1 = 4$ V, $v_2 = 8$ V. p: 8, 16, source -24 W *($\sum p = 0$ ✓)*



Worked: KCL with a node voltage: the circuit above

1. RESISTOR CURRENTS
$i_1 = \frac{v}{2}$, $i_2 = \frac{v}{3}$ *(ground is 0 V)*

↓

2. KCL AT a
$6 = \frac{v}{2} + \frac{v}{3} = \frac{5v}{6}$

↓

3. SOLVE
$v = 7.2$ V; $i_1 = 3.6$ A, $i_2 = 2.4$ A *($3.6 + 2.4 = 6$ ✓)*

9. series and parallel resistors [book 2-3]

Formula	In words
$R_{eq} = R_1 + R_2 + \cdots$	**series**: resistances add
$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$	**parallel**: reciprocals add, then flip
$R_1 \parallel R_2 = \frac{R_1R_2}{R_1+R_2}$	two in parallel: (product) ÷ (sum)
$\frac{R_1R_2R_3}{R_1R_2 + R_2R_3 + R_1R_3}$	three in parallel
$G_{eq} = G_1 + G_2 + \cdots$	parallel conductances add

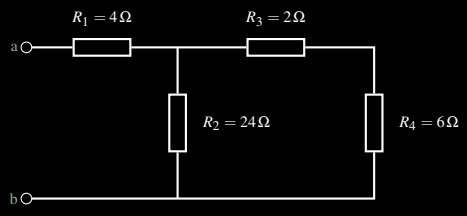
Series total > the biggest. Parallel total < the smallest. Two circuits are **equivalent** if they have the same i–v at their terminals.

Combination chart

Pattern	Formula	Example (Ω)
two in series	$R_1 + R_2$	$2 + 3 = 5$
three in series	$R_1 + R_2 + R_3$	$1 + 2 + 3 = 6$
two in parallel	$\frac{R_1R_2}{R_1+R_2}$	$6 \parallel 3 = \frac{18}{9} = 2$
two equal, parallel	$\frac{R}{2}$	$10 \parallel 10 = 5$
N equal, parallel	$\frac{R}{N}$	four 100s: $\frac{100}{4} = 25$
three in parallel (2, 3, 6)	$\frac{1}{\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}}$	$\frac{1}{0.5 + 0.333 + 0.167} = 1$
series, then parallel	$(R_1 + R_2) \parallel R_3$	$(2+4) \parallel 3 = 6 \parallel 3 = 2$
parallel, then series	$(R_1 \parallel R_2) + R_3$	$(6 \parallel 3) + 4 = 2 + 4 = 6$
two series pairs, parallel	$(R_1 + R_2) \parallel (R_3 + R_4)$	$(1+5) \parallel (4+2) = 6 \parallel 6 = 3$
wire across (short)	$R \parallel 0 = 0$	$6 \parallel 0 = 0$
open branch in parallel	$R \parallel \infty = R$	$6 \parallel \infty = 6$
open in series	$R + \infty = \infty$	no current flows
missing R, series	$R = R_{eq} - R_1$	$1000 - 330 = 670$
missing R, parallel	$R = \frac{R_{eq}R_1}{R_1 - R_{eq}}$	$R \parallel 12 = 4$: $\frac{4\cdot 12}{12-4} = 6$

How to reduce: label the nodes. Merge nodes joined by wires or shorts. Find a clear series pair or parallel pair. Replace it. Redraw. Repeat, starting

farthest from the terminals.



Worked: reduce the ladder above to R_{ab}

1. FAR END: $R_3 + R_4$ (SERIES)
$2 + 6 = 8\,\Omega$

↓

2. WITH R_2 (PARALLEL)
$24 \parallel 8 = \dfrac{192}{32} = 6\,\Omega$

↓

3. WITH R_1 (SERIES)
$R_{ab} = 4 + 6 = 10\,\Omega$

Worked: reduce $R_{eq} = 2 + \left[12 \parallel (4 + 6 \parallel 3)\right]\ \Omega$, inside out

1. $6 \parallel 3$
$\dfrac{18}{9} = 2\,\Omega$

↓

2. $4 + 2$
$6\,\Omega$

↓

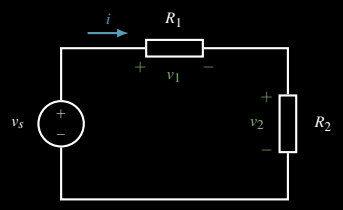
3. $12 \parallel 6$
$\dfrac{72}{18} = 4\,\Omega$

↓

4. $2 + 4$
$R_{eq} = 6\,\Omega$

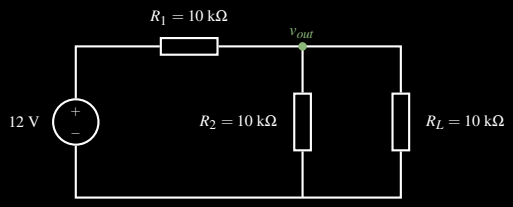
10. voltage and current dividers [book 2-3]

Formula	In words
$v_1 = \dfrac{R_1}{R_1 + R_2}\, v_s$	**voltage divider** (series): own R on top
$v_k = \dfrac{R_k}{R_1 + \cdots + R_N}\, v_s$	N series resistors
$R_2 = \dfrac{R_1\, v_2}{v_s - v_2}$	resistor for a target v_2
$i_1 = \dfrac{R_2}{R_1 + R_2}\, i_s = \dfrac{G_1}{G_{eq}}\, i_s$	**current divider** (parallel): other R on top, or own G
$i_k = \dfrac{R_{eq}}{R_k}\, i_s$	any number of parallel branches
$R_2 = \dfrac{R_1\,(i_s - i_2)}{i_2}$	resistor for a target i_2



Quick example	Work → answer
12 V, $R_1 = 2$ k, $R_2 = 4$ k	$v_1 = \frac{2}{6} \cdot 12 = 4$ V, $v_2 = 8$ V
12 V, $R_1 = 9$ k, want $v_2 = 3$ V	$R_2 = \frac{9 \cdot 3}{12 - 3} = 3$ kΩ
6 A into $3 \parallel 6\,\Omega$	$\frac{6}{9} \cdot 6 = 4$ A, $\frac{3}{9} \cdot 6 = 2$ A
12 A into 2, 3, 6 Ω ($R_{eq} = 1$)	$\frac{1}{2} \cdot 12 = 6$, $\frac{1}{3} \cdot 12 = 4$, $\frac{1}{6} \cdot 12 = 2$ A
10 A, $R_1 = 4\,\Omega$, want $i_2 = 2$ A	$R_2 = \frac{4(10-2)}{2} = 16\,\Omega$

A load across a divider changes it. Combine the load with the resistor it sits across first, then divide.



Worked: the loaded divider above

1. NO LOAD
$v_{out} = \dfrac{10}{20} \cdot 12 = 6$ V

↓

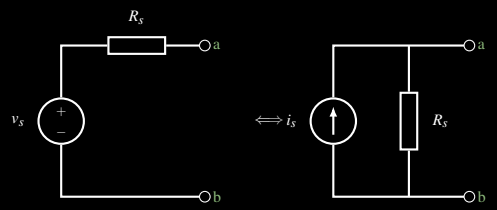
2. COMBINE THE LOAD FIRST
$R_2 \parallel R_L = 10 \parallel 10 = 5$ kΩ

↓

3. DIVIDE
$v_{out} = \dfrac{5}{10 + 5} \cdot 12 = 4$ V *(the load pulls it down)*

11. source transformation and combining sources [book 2-3]

A voltage source v_s in series with R_s acts exactly like a current source i_s in parallel with the same R_s.



Formula	In words
$i_s = \dfrac{v_s}{R_s}$	voltage source → current source, same R_s in parallel
$v_s = i_s R_s$	current source → voltage source, same R_s in series
arrow of i_s	points toward the + end of v_s
$v_{eq} = v_1 - v_2 + v_3$	voltage sources in series: add with signs
$i_{eq} = i_1 - i_2 + i_3$	current sources in parallel: add with directions

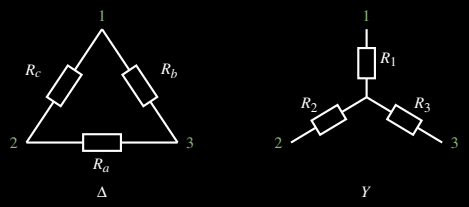
Quick example	Work → answer
12 V in series with 4 Ω	$i_s = 12/4 = 3$ A in parallel with 4 Ω
2 A in parallel with 5 Ω	$v_s = 2 \cdot 5 = 10$ V in series with 5 Ω
9 V and 3 V in series, opposing	6 V
2 A up, 5 A up, 1 A down, in parallel	$2 + 5 - 1 = 6$ A up
open-circuit voltage of a real source	no current, no drop on R_s: $V_{oc} = v_s$

Do not transform the element whose v or i you need. It disappears into the new source.

Unrealizable: different voltage sources in parallel, or different current sources in series. Each node pair has one voltage, each branch one current. This includes dependent sources.

12. wye–delta (Y–Δ, T–Π) [book 2-4]

For networks that are neither series nor parallel (a bridge). Both shapes connect the same three nodes 1, 2, 3. The book's labels: Y arms R_1, R_2, R_3 go to nodes 1, 2, 3. Δ side R_a is opposite node 1 (between 2 and 3), R_b opposite node 2, R_c opposite node 3.

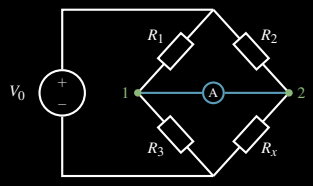


Formula	In words
$R_1 = \dfrac{R_b R_c}{R_a + R_b + R_c}$	$\Delta \to Y$: (Y arm) = (the two Δ sides touching its node, multiplied) ÷ (sum of all three)
$R_2 = \dfrac{R_a R_c}{\Sigma}, \quad R_3 = \dfrac{R_a R_b}{\Sigma}$	same pattern
$R_a = \dfrac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$	$Y \to \Delta$: (sum of pair products) ÷ (the Y arm at the node opposite)
$R_b = \dfrac{N}{R_2}, \quad R_c = \dfrac{N}{R_3}$	same pattern, N = the numerator above
$R_Y = \dfrac{R_\Delta}{3}, \quad R_\Delta = 3R_Y$	balanced (all equal)

Quick example	Work → answer
Δ: $R_a = 10, R_b = 20, R_c = 30$	$\Sigma = 60$: $R_1 = \frac{600}{60} = 10, R_2 = \frac{300}{60} = 5,$ $R_3 = \frac{200}{60} = 3.33$
(reverse) Y: 10, 5, 3.33	$N = 50 + 16.67 + 33.33 = 100$: $R_a = \frac{100}{10} = 10,$ $R_b = \frac{100}{5} = 20, R_c = \frac{100}{3.33} = 30$
balanced Δ, each 30 Ω	$R_Y = 10\,\Omega$

Other books swap the labels. Always match by *position*: a Y arm uses the two Δ sides that touch its node.

13. Wheatstone bridge and resistive sensors [book 2-5, 2-6.3]



Formula	In words
$V_1 = \dfrac{R_3}{R_1 + R_3} V_0,$ $V_2 = \dfrac{R_x}{R_2 + R_x} V_0$	each side is a voltage divider
$R_x = \dfrac{R_2}{R_1} R_3$	**balanced** ($I_a = 0$, $V_1 = V_2$): (unknown) from the other three
$V_{out} = \dfrac{V_0 \Delta R}{4R + 2\Delta R}$	sensor bridge: three arms R, one $R + \Delta R$ (exact)
$V_{out} \approx \dfrac{V_0}{4} \cdot \dfrac{\Delta R}{R}$	when $\Delta R \ll R$
$\dfrac{\Delta R}{R} = \dfrac{4V_{out}}{V_0}$	(relative change) from the reading

Piezoresistor (pressure sensor):

Formula	In words
$R = R_0(1 + \alpha P), \quad \Delta R = R_0 \alpha P$	P in N/m^2; $P > 0$ stretch, $P < 0$ squeeze
$P = \dfrac{F}{A}$	(pressure) = (force) ÷ (area)
$P = \dfrac{F L}{W H^2}$	cantilever: force F at the tip, length L, width W, thickness H
$V_{out} = \dfrac{V_0}{4} \alpha P = \dfrac{V_0}{4} \alpha \dfrac{F L}{W H^2}$	bridge output for a piezoresistor

Silicon: $\alpha \approx -1 \times 10^{-9}$ m^2/N (usually negative).

Quick example	Work → answer
$R_1 = 100, R_2 = 200$, balance at $R_3 = 150\,\Omega$	$R_x = \frac{200}{100} \cdot 150 = 300\,\Omega$
$V_0 = 5$ V, $\Delta R/R = 0.01$	$\frac{5}{4} \cdot 0.01 = 12.5$ mV (exact: 12.4 mV)
$V_0 = 4$ V reads 2 mV	$\frac{\Delta R}{R} = \frac{4 \cdot 0.002}{4} = 0.002$
cantilever $F = 10^{-3}$ N, $L = 1$ mm, $W = 0.1$ mm, $H = 10\,\mu$m	$P = \frac{10^{-3} \cdot 10^{-3}}{10^{-4} \cdot 10^{-10}} = 10^8$ N/m^2
same, $V_0 = 1$ V, $\alpha = -10^{-9}$	$V_{out} = \frac{1}{4}(-10^{-9})(10^8) = -25$ mV

14. nonlinear elements: fuse, diode, LED [book 2-1.4, 2-6]

- **Fuse**: a wire that opens when the current passes its rating. A short at the load draws $I = V_s/R_s$, which blows it.
- **Ideal diode**: a one-way valve. Forward ($V_D > 0$): a short. Reverse ($V_D < 0$): an open, $I = 0$.
- **Real diode model**: off until V_D reaches V_F, then a drop of V_F (plus a small R_D, often ignored). Silicon: $V_F \approx 0.7$ V.
- **LED**: a diode that lights. On fully at about 20 mA. Red $V_F \approx 1.6$ V, green or blue ≈ 3.3 V. Curve: $I \approx aV_D^2$ ($V_D > 0$).

Formula	In words
$I = \dfrac{V_s - V_F}{R}$	forward diode in series with R
$R = \dfrac{V_s - V_F}{I}$	resistor to set an LED current
$R_D \approx \dfrac{\Delta V_D}{\Delta I}$	slope of the curve above V_F

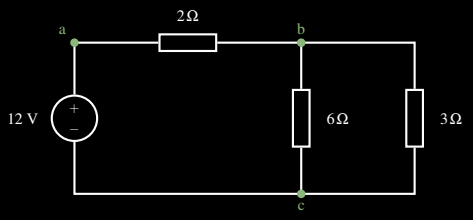
Diode procedure: (1) assume forward bias, (2) compute I, (3) if $I > 0$, done; if not, the diode is off and $I = 0$. For a square or triangle source, do each half of the cycle separately.

Quick example	Work → answer
5 V, 1 kΩ, Si diode forward	$I = \frac{5-0.7}{1000} = 4.3$ mA > 0 ✓
same source reversed (−5 V)	$\frac{-5-0.7}{1000} < 0 \Rightarrow$ off, $I = 0$
red LED at 20 mA from 5 V	$R = \frac{5-1.6}{0.02} = 170\,\Omega$

***I–V* slopes** [book Fig. 2-38]: resistor $1/R$, ideal voltage source vertical (∞), ideal current source flat (0).

15. putting it together: one full circuit

Reduce, find the source current, then expand back out with Ohm's law, KVL, and KCL.



Worked: find every current, voltage, and power

1. COMBINE: $R_2 \parallel R_3$, THEN $+R_1$
$6 \parallel 3 = 2\,\Omega$; $R_{eq} = 2 + 2 = 4\,\Omega$

↓

2. SOURCE CURRENT (OHM ON R_{eq})
$i = \dfrac{12}{4} = 3$ A

↓

3. VOLTAGE ON R_1, THEN ON THE PAIR (KVL)
$v_1 = 3 \cdot 2 = 6$ V; $v_{bc} = 12 - 6 = 6$ V

↓

4. BRANCH CURRENTS (OHM)
$i_2 = 1$ A, $i_3 = 2$ A *(KCL: $1 + 2 = 3$ ✓)*

↓

5. POWER
$p_1 = 18, p_2 = 6, p_3 = 12$ W. Total 36 W = $12 \cdot 3$ from the source ✓